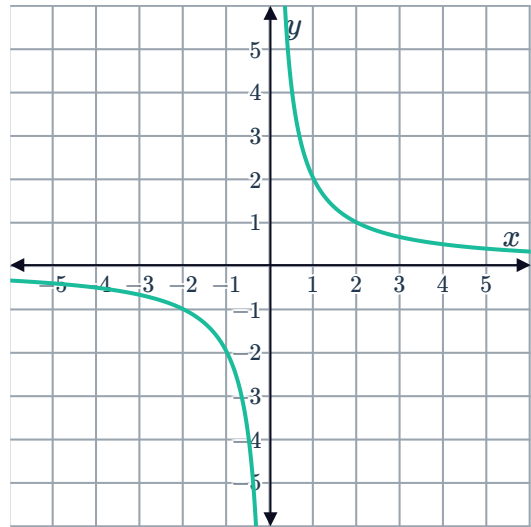


Key features of a hyperbola



1 Consider the graph of $y = \frac{2}{x}$.

- a For $x > 0$, as x increases, what does y approach?
- b For $x > 0$, as x approaches 0, what does y approach?
- c State the equation of the vertical asymptote.
- d State the equation of the horizontal asymptote.
- e The graph has two axes of symmetry. State their equations.



2 Consider the function $y = -\frac{12}{x}$.

- a When $x = 3$, what is the value of y ?
- b If x is a positive value, is the corresponding y value positive or negative?
- c If x is a negative value, is the corresponding y value positive or negative?
- d In which quadrants does the graph of $y = \frac{-12}{x}$ lie?

3 Consider the function $y = -\frac{5}{x}$.

- a For what value of x is the function undefined?
- b As x approaches 0 from the positive side, what does y approach?
- c As x approaches 0 from the negative side, what does y approach?
- d As x approaches ∞ , what does y approach?
- e As x approaches $-\infty$, what does y approach?

4 Consider the function $y = \frac{2}{x}$.



a Complete the table of values.

x	-2	-1	$-\frac{1}{2}$	$-\frac{1}{10}$	$-\frac{1}{100}$	$\frac{1}{100}$	$\frac{1}{10}$	$\frac{1}{2}$	1
y									

b For what value of x is the function undefined?

c Rewrite the equation to make x the subject.

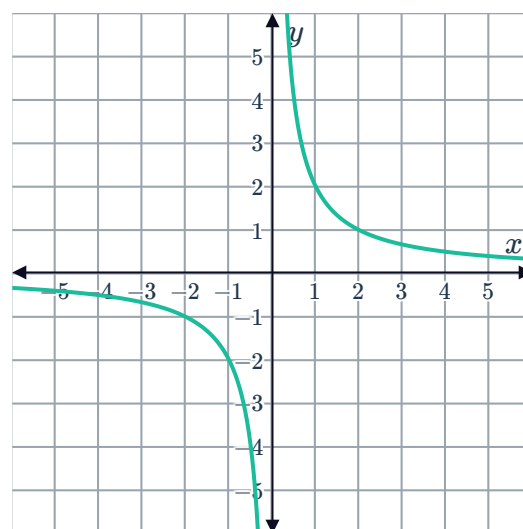
d For what value of y is the function undefined?

e Below is the graph of $y = \frac{2}{x}$.

Complete the following statement:

As x approaches from the right, the function value approaches ∞ . As x approaches from the left, the function value approaches $-\infty$.

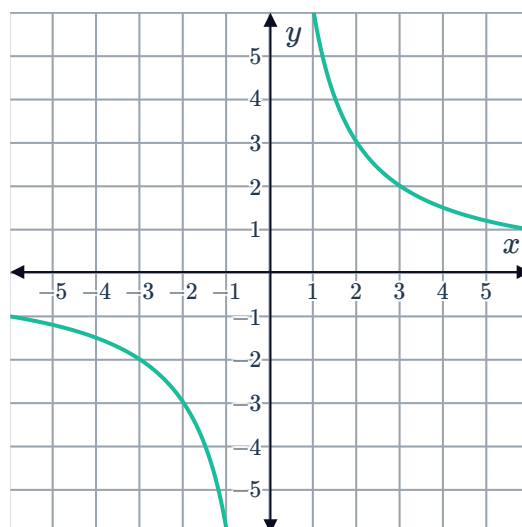
As x approaches ∞ and $-\infty$, y approaches . This is called the limiting value of the function.



5 Consider the hyperbola shown:

Complete the statement:

"Every point (x, y) on the hyperbola is such that $xy = \text{input}$."

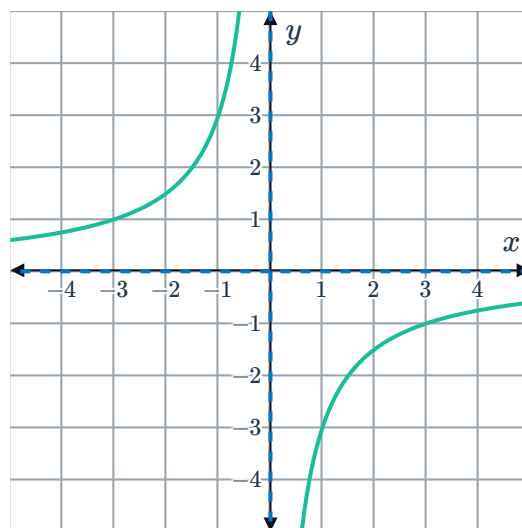


6 Determine whether the following relations  can be modelled by a function of the form $xy = a$?

- a The relationship between the number of people working on a job and how long it will take to complete the job.
- b The relationship between the number of sales and the amount of revenue.
- c The relationship between height and weight.

7 Consider the function graphed below:

- a What is the equation of the vertical asymptote?
- b What is the equation of the horizontal asymptote?
- c State the domain of the function.
- d State the range of the function.



8 For functions of the form $f(x) = \frac{k}{x}$, where k is a constant, the domain is always $(-\infty, 0) \cup (0, \infty)$. What does this mean for the function when $x = 0$?

9 Consider the function $y = \frac{1}{x}$.

- a For what value of x is the function undefined?
- b What is the domain of $y = \frac{1}{x}$?
- c For what value of y is the function underfined?
- d What is the range of $y = \frac{1}{x}$?

10 Patricia determined the range of the function $y = \frac{2}{x}$ is $(-\infty, 0) \cup (0, \infty)$.

By filling in the gaps below, complete her reasoning.

Looking at $y = \frac{2}{x}$, notice that the numerator is non-zero, and so $\frac{2}{x}$ can never be equal to .

Another approach is to rearrange the equation and make x the subject. Then we get $x = \text{$. In this form we can see that the denominator cannot be equal to .



11 Consider the functions $y = \frac{4}{x}$ and $y = \frac{2}{x}$. 

- a When $x = 2$, what is the value of y if $y = \frac{4}{x}$?
- b When $x = 2$, what is the value of y if $y = \frac{2}{x}$?
- c Which graph lies further away from the axes?
- d For hyperbolas of the form $y = \frac{k}{x}$, as the k value increases, what happens to the graph?

Graphs of hyperbolas

12 Consider the function $y = \frac{6}{x}$.

- a Complete the following tables of values:

x	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4
y					

- b Plot the points from the table of values on a cartesian plane.

13 Ursula wants to sketch the graph of $y = \frac{7}{x}$, but knows that it will look similar to many other hyperbolas.

What can she do to the graph to show that it is the hyperbola $y = \frac{7}{x}$, rather than any other hyperbola of the form $y = \frac{k}{x}$?

14 For each of the following functions:

- i Complete a table of values of the form:

x	-2	-1	$-\frac{1}{2}$	$\frac{1}{2}$	1	2
y						

- ii Sketch the graph of the function.
- iii State which quadrants the function lies in.

a $y = \frac{2}{x}$

b $y = -\frac{1}{x}$

c $y = -\frac{1}{4x}$

Applications



- 15** The relationship between the current, C , (in amperes) and resistance, R , (in ohms) in an electrical circuit is given by:

$$C = \frac{240}{R}$$

where the voltage provided to the circuit is 240 V.

- Graph this function for $0 \leq R \leq 50$.
- What happens to the current as the resistance increases?

- 16** The rent, electricity, telephone bill and other expenses for a flat costs a total of \$490 per week. These expenses are shared equally between the tenants of the flat.

- How much will each occupant pay in dollars if the flat is shared by two people?
- Let the number of occupants be x , and the cost paid by each occupant be y .

Write a formula that relates the two pronumerals.

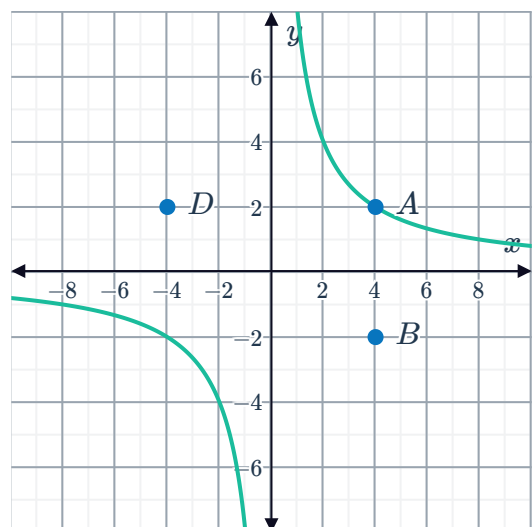
- Use this equation to fill in the following table of values.

x	1	2	3	4	5	6
y						

- Sketch the curve from your equation.
- What type of relationship exists between x and y ?

- 17** Consider the hyperbola that has been graphed. Points $A(4, 2)$, B , $C(-4, k)$ and D form the vertices of a rectangle.

- Find the value of k .
- Hence find the area of the rectangle with vertices $ABCD$.



- 18** Boyle's law describes the relationship between  pressure and volume of a gas of fixed mass under constant temperature. The pressure for a particular gas can be found using

$$P = \frac{6000}{V}$$

where P has units kg/cm^2 and V has units cm^3 .

- a** Graph the relationship $P = \frac{6000}{V}$ for $0 \leq V \leq 2000$.
 - b** What is the pressure if the volume is 1 cm^3 ?
 - c** What happens to the pressure as the volume increases?
- 19** The time it takes a commuter to travel 100 km depends on how fast they are going. We can write this using the equation $t = \frac{100}{S}$ where S is the speed in km/h and t is the time taken in hours.
- a** Graph the relationship $t = \frac{100}{S}$.
 - b** What is the time taken if the speed travelled is $10 \text{ km}/\text{h}$?
 - c** What is the time taken if the speed travelled is $50 \text{ km}/\text{h}$?
 - d** If we want the travel time to decrease, what must happen to the speed of travel?